1 Planning (15 marks)

Defining the problem (3 marks)

P t is the independent variable and I (or amplitude of reflected signal) is the dependent variable, or vary t and measure I (or amplitude of reflected signal). [1]

P Keep distance from the wall/foam to the speaker/microphone constant. [1]

P Keep the amplitude or intensity I_0 of the sound before reflection constant. [1]

Methods of data collection (5 marks)

M Labelled diagram of workable experiment including speaker, microphone/sound detector, foam and wall. [1]

M Signal generator/a.c. power supply connected to speaker. [1]

M Microphone connected to oscilloscope or sound (intensity) meter. [1]

M Measure the thickness with a rule/micrometer/vernier calipers. [1]

M Method to determine the density; $\rho = m/V$. [1]

Method of analysis (2 marks)

A Plot a graph of ln I against t.
(Allow log I against t and lg I against t graphs.) [1]

A $\alpha = -\text{gradient}/\rho$ (must be consistent with graph plotted) [1]

Safety considerations (1 mark)

S Precaution linked to loud sounds, e.g. use ear plugs/muffs/defenders.
Allow switch off sound source to prevent damage to ears. [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Keep the frequency constant
2 Carry out experiment in a quiet room/no other sources of sound
3 Method to keep angles constant/positions of speaker and microphone constant.
4 Method and explanation to detect reflected sound from foam only, e.g. barrier, tube or method to avoid reflections
5 Method to determine mass, e.g. use scales/balance and method to determine volume
6 Relationship is valid if the graph is a straight line (ignore reference to y-intercept)
7 Method to check that emitted sound I_0 is constant or method to check y-intercept is ln I_0.
8 Intensity is proportional to the amplitude^2.

Do not allow vague computer methods.

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